

illuminati - 2019

Solutions to Advanced Physics Assignment-4B

Modern Physics & Optics

1.(ACD)

In radiation A incident beam passes from which some of the wavelengths will be absorbed by the hydrogen gas hence option (A) is correct. Due to absorption of ultraviolet radiation electrons get excited and during reverse transition different spectral series of hydrogen atom are emitted in radiation B hence option (C) and (D) are correct.

2.(AC) For inelastic collisions the maximum possible energy loss in a collision should be more than the minimum energy required for excitation of any of the colliding atoms which in this case is 10.2 eV for hydrogen atom. As the hydrogen atom and neutron are of same mass, maximum kinetic energy loss can be the half of initial kinetic energy of the colliding particle hence in above case initial kinetic energy must be more than or equal to 20.4 eV for inelastic collision hence options (A) and (C) are correct.

3. By momentum conservation $m_{Li}v_{Li} = m_{He}v_{He} \dots (i)$

KE of Li nucleus is

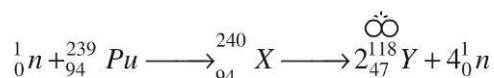
$$k_{Li} = \frac{1}{2}m_{Li}v_{Li}^2 = \frac{1}{2}m_{Li} \left[\frac{m_{He}v_{He}}{m_{Li}} \right]^2 = \frac{1}{2} \frac{m_{He}^2 \cdot v_{He}^2}{m_{Li}} = \frac{1}{2} \frac{(4.0026 \times 1.66 \times 10^{-27})^2 \times (9.3 \times 10^6)^2}{(7.0160 \times 1.66 \times 10^{-27})}$$

$$= 1.65 \times 10^{-13} J = 1.03 MeV$$

Q-value of reaction is

$$Q = k_{Li} + k_{He} = 1.03 MeV + \frac{1}{2} \frac{(4.0026 \times 1.66 \times 10^{-27}) (8.3 \times 10^6)^2}{1.6 \times 10^{-19}} = 1.03 MeV + 1.80 MeV$$

4. Nuclear reaction



Charge on each Y is $q = 4.7 \times 1.6 \times 10^{-19} = 7.52 \times 10^{-18} C$

Radius of each Y is $R = R_0(A)^{1/3} = 1.1 \times 10^{-15} (118)^{1/3} = 5.395 \times 10^{-15} m$

At the time 2Y are produced thin separation is $l = 2R = 10.79 \times 10^{-15} m$

Force between these nuclei of Y is

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{l^2} = (9 \times 10^9) \times \frac{(7.52 \times 10^{-18})^2}{(10.79 \times 10^{-15})^2} = 4.37 \times 10^3 N = 4.37 kN$$

5. Nuclear reaction ${}^{57}_{C_0} \xrightarrow[T=270 \text{ days}]{\beta^+} {}^{57}_{Fe^+} \xrightarrow[T_2=10^{-8} \text{ sec}]{r} {}^{57}_{Fe} + r$

Here overall half-life of r -emission can be taken as 270 days as just after one β^+ decay it is suddenly following by r -emission.

$$\Rightarrow \text{By RA decay law } A_{c\beta} = A_{cr} = A_{c0} (2)^{-t/T}$$

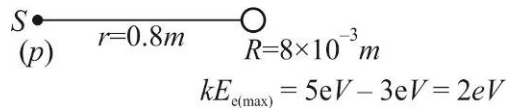
$$(1.25 \times 10^5) = (5 \times 10^9) (2)^{-t/270}$$

$$2^{-2} = 2^{-t/270}$$

$$t = 2 \times 270 = 540 \text{ days.}$$

6. Energy incident on sphere is $E = \frac{P}{4\pi r^2} \times \pi R^2 J/s$

Number of photons incident on sphere $N = \frac{E}{(5eV)} \text{ ph/sec}$



$$kE_{e(\max)} = 5eV - 3eV = 2eV$$

Number of s ejected/sec after time t $q_t = n_e \cdot e \cdot t$

PEE Stops from sphere when its potential rises to 2V $\Rightarrow \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R} = 2$

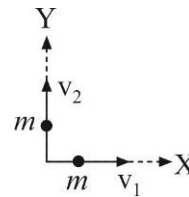
$$\frac{1}{4\pi\epsilon_0 R} \cdot \frac{P R e t}{4r^2 (5eV) 10^6} = 2$$

$$t = \frac{2 \times 4 \times (0.8)^2 \times 5e \times 10^6}{3.2 \times 10^{-3} \times 8 \times 10^{-3} \times e \times 9 \times 10^9} \Rightarrow t = 111 \text{ sec}$$

7. Velocity of COM of particles $\vec{V}_c = \frac{mv_1\hat{i} + mv_2\hat{j}}{2m} = \frac{v_1\hat{i} + v_2\hat{j}}{2}$

Volume of first part w.r.t. COM $\vec{V}_{ic} = \vec{v}_1 - \vec{v}_c = \frac{v_1\hat{i} - v_2\hat{j}}{2}$

Volume of sec part w.r.t to COM $\vec{v}_{2c} = \vec{v}_2 - \vec{v}_c = \frac{v_2\hat{j} - v_1\hat{i}}{2}$



De-broglie w/L of particle of w.r.t to COM

$$\lambda_{1c} = \frac{h}{mv_{1c}} = \frac{2h}{m\sqrt{v_1^2 + v_2^2}} = \frac{2h}{m\sqrt{\left(\frac{h}{m\lambda_1}\right)^2 + \left(\frac{h}{m\lambda_2}\right)^2}} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$$

$$\lambda_{2c} = \frac{h}{mv_{2c}} = \frac{2h}{m\sqrt{v_1^2 + v_2^2}} = \frac{2h}{m\sqrt{(h/m\lambda_1)^2 + (h/m\lambda_2)^2}} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$$

8. Energy of incident radiation $E = \frac{12431}{4000} = 3.1eV$

Max KE of ejected electrons $KE_{\max} = E - hv_{+h} = 3.1 - 1.9 = 1.2eV$

Now these e^- combine with He^{+2} produce He^+ in $n = 5$ state (fourth excited state)

$$\Rightarrow \text{Energy released in } \Delta E = 1.2 + \frac{13.6z^2}{h^2} = 1.2 + \frac{13.6 \times 4}{25} = 1.2 + 2.18 = 3.38eV$$

Energies of orbitals of He^+ are $E_1 = -13.6 \times 2^2 = -54.4eV$

$$E_2 = -\frac{13.6 \times 2^2}{2^2} = -13.6eV \quad E_3 = -\frac{13.6 \times 2^2}{3^2} = -6.04eV$$

$$E_4 = -\frac{13.6 \times 2^2}{4^2} = -3.4eV$$

$$E_5 = -\frac{13.6 \times 2^2}{5^2} = -2.18 \text{ eV}$$

Energies in range from 2 to 4 eV are $\Delta E_{5 \rightarrow 3} = (-2.18) - (-6.04) = 3.86 \text{ eV}$

$$\Delta E_{4 \rightarrow 2} = (-3.4) - (-6.04) = 2.64 \text{ eV}$$

$$\Delta E_i = 3.38 \text{ eV}$$

9. For H-atom, rev frog of e^- in ground state in ($n = 1$) $f_1 = \frac{v}{2\pi r} = \frac{(2\pi k e^2 / h)}{2\pi(h^2 / 4\pi^2 k e^2 m_e)}$

$$f_1 = \frac{4\pi^2 k^2 e^4 m_e}{h^3}$$

For primeson, mass of e^- can be changed to reduced mass as $\mu_k = \frac{m_p(273m_e)}{m_p + 273m_e}$

The rev frog of prion is $f = \frac{4\pi^2 k^2 e^4 m_p(273m_e)}{h^3[m_p + 273m_e]}$

$$f = 1.77 \times 10^{18} \text{ sec}^{-1}$$

Number of rev in 10^{-8} s are $N = f \times \Delta t = 1.77 \times 10^{-18} \times 10^{-18} = 1.77 \times 10^{10} \text{ rev}$

10. First line of Balmer series corresponds to the transition $n_2 = 3$ to $n_1 = 2$

$$\text{For A } \Delta E_A = 13.6 Z_A^2 \left[\frac{1}{4} - \frac{1}{9} \right] = \frac{13.6 \times 5}{36} Z_A^2 \text{ eV}$$

$$\text{For B } \Delta E_B = 13.6 Z_B^2 \left[\frac{1}{4} - \frac{1}{9} \right] = \frac{13.6 \times 5}{36} Z_B^2 \text{ eV}$$

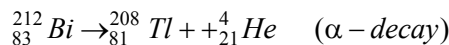
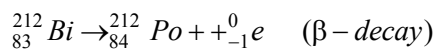
$$\text{It is given that } 5.567 = \Delta E_B - \Delta E_A = \frac{13.6 \times 5}{36} (Z_B^2 - Z_A^2) \Rightarrow Z_B^2 - Z_A^2 = 3 \quad \dots\dots\dots(1)$$

It is also given that $2m_B v = 2(2m_A v)$ [We can use $m_A = 2Z_A$ & $2Z_B$]

$$Z_B = 2Z_A \quad \dots\dots\dots(2)$$

From equation (1) $4Z_A^2 - Z_A^2 = 3 \Rightarrow Z_A = 1, Z_B = 2$ [$A \rightarrow 1$ H (Hydrogen), $B \rightarrow 2$ (Helium)]

11. The nuclear reaction for decay are



After 1 Hour the number of atoms of Bi will reduce to half $\Rightarrow$ at $t = 1$ hour 0.5 gm of Bi will decay.

At $t = 1$ Hour masses in the mixture are

$$m_{\text{Bi}} = 0.5 \text{ gm}$$

$$m_{P_0} = \frac{13}{20} \times 0.5 = 0.325 \text{ gm}$$

$$m_{T_l} = \frac{7}{20} \times 0.5 = 0.175 \text{ gm}$$

12. RA series is ${}_{90}^{232}\text{Th} \rightarrow {}_{82}^{208}\text{Pb}$

$\Rightarrow$ Mass no difference = $232 - 208 = 24$ [only due to emission of α - part]

$\Rightarrow$ Number of α - particles emitted by each $\text{Th} = \frac{24}{4} = 6\alpha$ part/decay

$$\text{No. of He atoms in sample } N_{\alpha} = \frac{PV}{KT} = \frac{1.013 \times 10^5 \times 8 \times 10^{-5} \times 10^{-6}}{1.38 \times 10^{-23} \times 273} = 2.151 \times 10^{15} \text{ atom}$$

$$\text{No. of } ^{232}\text{Th} \text{ decay in total } \frac{N_{\alpha}}{6} = 0.3585 \times 10^{15} \text{ decays}$$

$$\text{Total No. of } ^{232}\text{Th} \text{ nuclei in sample } N_0 = \frac{5 \times 10^{-7} \times 6.023 \times 10^{23}}{232} = 1.298 \times 10^{15} \text{ Nu}$$

$$\text{No. of } ^{232}\text{Th} \text{ left at present are } N = N_0 - \frac{N_{\alpha}}{6} = 0.9395 \times 10^{15} \text{ Nu}$$

$$\text{By RA decay law } N = N_0(2)^{-t/T} \Rightarrow t = -\frac{T \log(7.2 \times 10^{-1})}{0.301} = \frac{1.39 \times 10^{10} \times 0.142}{0.301} = 6.59 \times 10^9 \text{ years}$$

13. For stationary wave formation $d_1 = N \frac{\lambda}{2} = 2A^\circ$ (1)

For next higher separation $d_2 = (NH)\lambda / 2 = 2.5A^\circ$ (2)

Equation (2) – (1)

$$\Rightarrow \lambda / 2 = 0.5A^\circ \Rightarrow \lambda = 1A^\circ = 10^{-10} \text{ m}$$

By De-Broglie's hypothesis, momentum of e^- $P = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{10^{-10}}$
 $P = 6.63 \times 10^{-24} \text{ N.S}$

$$\text{Energy of } e^- \text{ (in eV)} E = \frac{P^2}{2m} = \frac{(6.63 \times 10^{-24})^2}{2 \times 9.1 \times 10^{-31}} \times \frac{1}{1.6 \times 10^{-19}} = 150.6 \text{ eV}$$

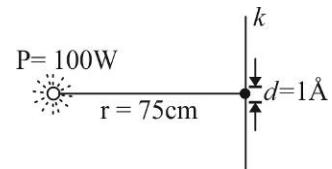
For s/w to be obtained min separation between two nodes is $= \frac{\lambda}{2} = 0.5A^\circ$

14. Light intensity at surface on its centre is $I = \frac{P}{4\pi r^2} \text{ w/m}^2$

Power received by each atom on surface at centre is P_R

$$P_R = I \cdot \frac{\lambda d^2}{4} = 100 \times \frac{5}{100} \times \frac{1}{4 \times (0.75)} \times \frac{xn(10^{-10})^2}{4}$$

$$t = \frac{2 \times 1.6 \times 10^{-19} \times 4 \times (0.75)^2}{5 \times (0.5 \times 10^{-10})^2} = 57.6 \text{ sec}$$



If in time t atom receives energy equal to 2eV-

$$\text{We use } P_R t = 2 \times 1.6 \times 10^{-19}$$

As $d \ll r$, no change in time will be there as energy received by atom remain same even if it is considered as a sphere.

15. If v is the max speed of ejected electrons, here to move in circular path we use

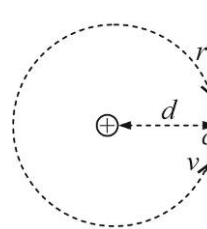
$$\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{d^2} = \frac{mv^2}{d} \text{(1)}$$

We also have

$$\frac{hc}{\lambda} = S + \frac{1}{2}mv^2$$

$$\frac{hc}{\lambda} = S + \frac{1}{2} \left(\frac{e^2}{4\pi\epsilon_0 d} \right)$$

$$\lambda = \frac{8x\epsilon_0 hcd}{e^2 + 8\pi\epsilon_0 Sd}$$



16. ${}^9\text{Be}(\alpha, n){}^{12}\text{C}$

As no ext force is there

So momentum of system remain conserved

$$x \rightarrow m_\alpha v_\alpha = m_c v_c \cos \theta \quad \dots\dots\dots(1)$$

$$y \rightarrow m_n v_n = m_c v_c \sin \theta \quad \dots\dots\dots(2)$$

Square & Added (1) & (2)

$$m_\alpha^2 v_\alpha^2 + m_n^2 v_n^2 = m_c^2 v_c^2 \quad \dots\dots\dots(3)$$

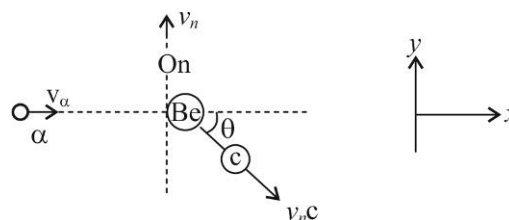
$$\text{By cons of energy } \frac{1}{2} m_\alpha v_\alpha^2 + \theta = \frac{1}{2} m_n v_n^2 + \frac{1}{2} m_c v_c^2 \quad \dots\dots\dots(4)$$

From Equation (3)

$$k + Q = k_n + \frac{1}{2} m / c \left[\frac{m_\alpha^2 v_\alpha^2 + m_n^2 v_n^2}{m_c^2} \right]$$

$$k + Q = k_n + k \left(\frac{m_n}{m_c} + k_n \left(\frac{m_n}{m_c} \right) \right) \left[\frac{m_\alpha^2 v_\alpha^2 + m_n^2 v_n^2}{m_c^2} \right]$$

$$k_n = \frac{Q + k \left(1 - \frac{m_n}{m_c} \right)}{\left(1 + \frac{m_n}{m_c} \right)}$$



17. Q-value of reaction

$$\theta = \delta m \times 931.5 \text{ MeV} = (228.028726 - 224.020196 - 4.00260) \times 931.5 \text{ MeV} = 5.524 \text{ MeV}$$

In first step of reaction (α – emission) energy released is $E_T = 5.524 - 0.217 = 5.307 \text{ MeV}$

$$\text{K.E. of } \alpha \text{ – particle emitted can be given } E_\alpha = E_r \left(\frac{A-4}{A} \right)$$

$$E_\alpha = 5.307 \left(\frac{224}{228} \right) \text{ MeV} = 5.214 \text{ MeV}$$

18. If E_α is the energy of α – emission then Q-value of reaction can be given as

$$Q = E_\alpha \left(\frac{A}{A-4} \right) = 1.83 \times \frac{11}{7} \text{ MeV} = 2.876 \text{ MeV}$$

$$\text{Here we can use } Q = \Delta m \times 931.5 \text{ MeV} \Rightarrow \Delta m = \frac{2.876}{931.5} = 0.00308 \text{ amu}$$

$$\text{Mass defect of the reaction is } (m_B + m_n) - (m_{Li} + m_{He}) = 0.00308 \text{ amu}$$

$$10.01167 + 1.00894 - M_{Li} - 4.00386 = 0.00308 \text{ amu}$$

$$M_{Li} = 7.01675 - 0.00308 = 7.01366 \text{ amu}$$

19. Q-value of reaction is $Q = (M_N - M_c - 2M_e) \times 931.5 \text{ MeV}$

$$= (12.018613 - 12 - 2 \times 0.00055) \times 931.5 \text{ MeV} = 16.3134 \text{ MeV}$$

In step = I, KE of e^+ will be maximum when $E_\gamma = 0$ & released energy is shared between

${}^{12}\text{C}^+$ & e^+ only. By momentum Gas.

$$16.3134 - 4.43 = \frac{1}{2} M_c V_c^2 + \frac{1}{2} m_e v_e^2 \text{ \& we also we } m_e v_e = M_c V_c$$

$$11.8834 = \frac{1}{2} M_c \left(\frac{m_e v_e}{M_c} \right)^2 + \frac{1}{2} m_e v_e^2 \quad \& \quad \text{we also we } v_c = \frac{m_e v_e}{M_c}$$

$$11.8834 = k_e \left[1 + \frac{m_e}{M_c} \right]$$

$$\text{Max. KE of } \beta^+ \text{ particle} \rightarrow k_e = \frac{11.8834 \times M_c}{m_e + M_c} = \frac{11.8834 \times 12}{12 + 0.00055} = 11.8828 \text{ MeV}$$

20. Q-value of reaction $Q = E_\alpha \left(\frac{A}{A-4} \right) = 5.30 \times \frac{210}{206} = 5.403 \text{ MeV}$

If daughter Nu after α -emission is γ^* , in second case the released energy is shared between α -particle & γ^*

$$\Rightarrow \text{released energy in second case is } Q_i' = E_\alpha \left(\frac{A}{A-4} \right) = 4.50 \times \frac{210}{206} = 4.587 \text{ MeV}$$

Difference of above two energies is absorbed by daughter Nu in its excitation.

Which is released in form of r -rod.

$$\text{Energy of gamma photon emitted } E_r = Q - Q_i' = 5.403 - 4.587 \text{ MeV} = 0.8156 \text{ MeV}$$

21. Area of light beam in which light power incident on black

$$S = ab \sin \theta + hb \cos \theta$$

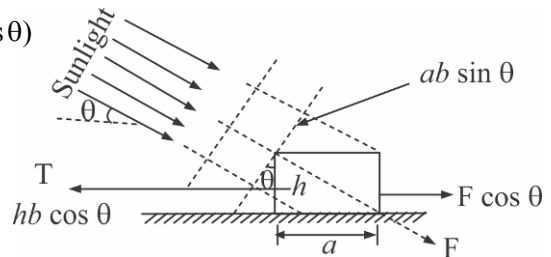
Power incident on block is $P = IS = Ib(a \sin \theta + h \cos \theta)$

Four on block due to abs of photons

$$F = \frac{P}{C} = \frac{Ib}{C} (a \sin \theta + h \cos \theta)$$

Tension in string $T = F \cos \theta$

$$T = \frac{Ib}{C} (a \sin \theta + h \cos \theta) \cos \theta$$



22. If $T \rightarrow T'$ as $E \propto T^4 \Rightarrow \frac{E}{E'} \left(\frac{T}{T'} \right)^4 \Rightarrow T' = (81)^{1/4} = 3T$

$$\text{By using Wein's displacement law } \lambda_m T = \lambda_m' T' \Rightarrow \lambda_m' = \frac{\lambda_m}{3} = 3000 \text{ \AA}$$

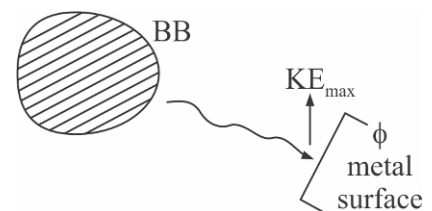
Energy of photons corresponding to peak spectral intensity is:

$$E = \frac{12431}{3000} = 4.14 \text{ eV}$$

$$\text{Given that } KE_{\max} = E_{H_3} - E_{H_2} = 1.89 \text{ eV} = eV_0$$

By equation of PEE, we use $E = Q + KE_{\max}$

$$Q = E - KE_{\max} = 4.14 - 1.89 = 2.25 \text{ eV}$$

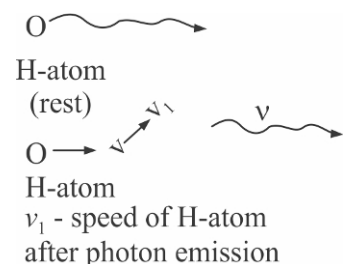


23. When atom is at rest $h\nu_0 = Rch \left[\frac{1}{m^2} - \frac{1}{n^2} \right]$

When atom is moving, we use

$$\frac{1}{2} mv^2 + Rch \left[\frac{1}{m^2} - \frac{1}{n^2} \right] = \frac{1}{2} mv_1^2 + h\nu \quad \dots(1)$$

$$\text{By cons. of momentum } mv = mv_1 + \frac{h\nu}{c} \quad \dots(2)$$



$$\Rightarrow v_1 = v - \frac{hv}{mc}$$

From equation (i)

$$\frac{1}{2}mv^2 + mv^2 + hv_2 = \frac{1}{2}m\left(v - \frac{hv}{mc}\right)^2 + hv$$

$$\frac{1}{2}mv^2 + mv_0 = \frac{1}{2}mv^2 + \frac{1}{2}m\left(\frac{hv}{mc}\right)^2 - \frac{mvhv}{mc} + hv$$

$$v\left(1 - \frac{v}{c}\right) = v_0 \Rightarrow v = v_0\left(1 - \frac{v}{c}\right)^{-1} = v_0\left(1 + \frac{v}{c}\right)$$

24. If atoms are excited to n^{th} level

$$\Rightarrow E_{\text{max.}} \rightarrow n \text{ to } i$$

$$52.224 = 13.6z^2 \left[\frac{1}{1^2} - \frac{1}{n^2} \right] \quad \dots(i)$$

$$\Rightarrow E_{\text{min.}} \rightarrow n \text{ to } n-1$$

$$1.224 = 13.6z^2 \left[\frac{1}{(n-1)^2} - \frac{1}{n^2} \right] \quad \dots(ii)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{52.224}{1.224} = \frac{(n^2-1)(n-1)^2}{2n-1}$$

$$\text{from equation (i)} \quad 52.224 = 13.6z^2 \left[1 - \frac{1}{25} \right]$$

$$52.224 = 13.6z^2 \times \frac{24}{25} \Rightarrow z = 2$$

$$\text{gas} \rightarrow \text{He}^+$$

$$\Rightarrow 42.67(2n-1) = (n^2-1)(n-1)^2$$

This is a fourth order equation so here we check for which integral value of n times equation will satisfy using $n = 3, 4, 5, 6 \dots$

We find at $n = 5$ above equation holds.

$$\Rightarrow n = 5$$

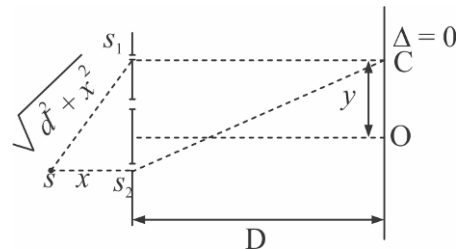
25. If central maxima is located a distance y from screen centre O , we use :

$$\sqrt{d^2 + x^2} - x = \frac{dy}{D}$$

$$y = \frac{D}{d} \left[x \sqrt{1 + \frac{d^2}{x^2}} - x \right] \quad v = \frac{-dx}{dt}$$

$$= \frac{D}{d} \left[x \left(1 + \frac{d^2}{2x^2} \right) - x \right] = \frac{Dd}{2x}$$

$$v_c = \frac{dy}{dt} = \frac{Dd}{2x^2} \cdot \frac{dx}{dt} = \frac{1 \times 0.01}{2(0.5)^2} \times 10^{-3} = 0.02 \text{ mm/s}$$



26. $I = 0$ at pts of destructive $\Delta = (2N + 1)\lambda / 2$ interference

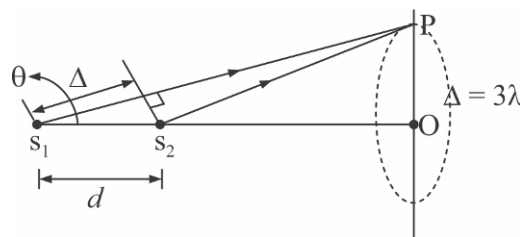
Path diff. at point P is $\Delta = d \cos \theta = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}$

$$d \cos \theta_1 = \frac{\lambda}{2} \Rightarrow \theta_1 = \cos^{-1} \left(\frac{\lambda}{2d} \right) = \cos^{-1} \left(\frac{1}{6} \right)$$

$$d \cos \theta_2 = \frac{3\lambda}{2} \Rightarrow \theta_2 = \cos^{-1} \left(\frac{3\lambda}{2d} \right) = \cos^{-1} \left(\frac{1}{2} \right)$$

$$d \cos \theta_3 = \frac{5\lambda}{2} \Rightarrow \theta_3 = \cos^{-1} \left(\frac{5\lambda}{2d} \right) = \cos^{-1} \left(\frac{5}{6} \right)$$

$\Rightarrow$ Three dark circular fringes are obtained on screen (wall).



27. Power through apertures A and B are

$$P_A = I \cdot A_A = \left(\frac{10}{\pi} \right) \times \pi (0.001)^2 = 10^{-5} \text{ W}$$

$$P_B = I \cdot A_B = \left(\frac{10}{\pi} \right) \times \pi (0.002)^2 = 4 \times 10^{-5} \text{ W}$$

Due to the film, path diff in path of light beams of D is $\Delta = t(\mu - 1)$

$$\text{Phase difference in waves at } D \text{ is } Q = \frac{2\pi}{\lambda} t(\mu - 1) = \frac{2\pi}{6 \times 10^{-7}} \times 2 \times 10^{-7} \times 0.5 = \frac{\pi}{3}$$

Resulting power at D (due to interference)

$$P_D = (P_A + P_B + 2\sqrt{P_A P_B} \cos \theta) \times \frac{10}{100}$$

$$P_D = 0.1 \times (10^{-5} + 4 \times 10^{-5} + 2 \times 2 \times 10^{-5} \times 1/2) = 7 \times 10^{-6} \text{ W}$$

28. By Snell's law $\sin 60^\circ = \sqrt{3} \sin r \Rightarrow \sin r = \frac{1}{2}$ or $r = 30^\circ$

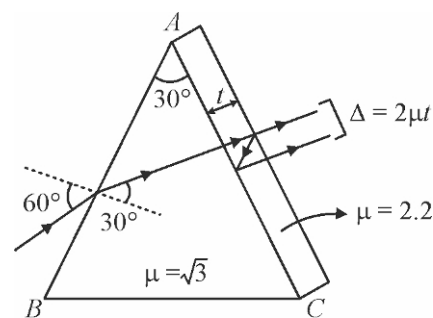
Path difference in the two light beams energy from face AC of prism is

$$\Delta = 2\mu t$$

For minimum intensity of light emerging from face AC we use

$$\Delta = 2\mu t = \frac{\lambda}{2} \text{ [for destructive interference]}$$

$$t = \frac{\lambda}{4\mu} = \frac{6.6 \times 10^{-7}}{4 \times 2.2} = 750 \text{ \AA}$$



29. In figure shown distances $SS_1 = SS_2 = 2a \cos \theta$

Separation between S_1 & S_2 is –

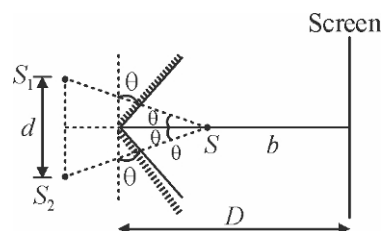
$$d = 2(2a \cos \theta) \sin \theta = 2a \sin 2\theta \approx 4a\theta \quad [\text{as } \theta \text{ is small}]$$

Separation between shift plane ($S_1 S_2$) and screen

$$D = (2a \cos \theta) \cos \theta + b \quad [\text{for small } \theta \cos \theta \approx 1]$$

$$D = 2a + b$$

$$\text{Fringe width in interference pattern } \beta = \frac{\lambda D}{d} = \frac{\lambda(2a + b)}{4a\theta}$$



30. If each slit produces intensity I_0 on screen $\Rightarrow I_{\max} = 4I_0$ path diff in lights from S_1 to S_2 to screen centre C-

$$\Delta_c = t_1 \left(\frac{\mu_s}{\mu_o} - 1 \right) - t_2 \left(\frac{\mu_s}{\mu_o} - 1 \right) = (t_1 - t_2) \left(\frac{\mu_s}{\mu_o} - 1 \right) = 1.25 \times 10^{-6} \left(\frac{1.4}{4/3} - 1 \right) = \frac{1}{16} \times 10^{-6} m$$

$$\text{Phase difference of C is } \phi = \frac{2\pi}{(\lambda / \mu_o)} \cdot \Delta = \frac{2\pi}{5 \times 10^{-7}} \cdot \left(\frac{4}{3} \right) \cdot \frac{1}{16} \times 10^{-6} = \frac{\pi}{3}$$

Intensity at screen centre is

$$I_C = 4I_0 \cos^2(\phi / 2) = 4I_0 \cos^2(\pi / 6) = 3I_0$$

$$\text{Ratio } \frac{I_C}{I_{\max}} = \frac{3I_0}{4I_0} = \frac{3}{4}$$

31. For TIR at face AC, it should be for $\lambda = 4000 \text{ \AA}$ such that is $\mu = \frac{1}{\sin \theta_c} = \frac{5}{4} = 1.25$ from the given relation

$$1.25 = 1.2 + \frac{b}{(4 \times 10^{-7})^2} \Rightarrow b = 8 \times 10^{-15}$$

$$\text{Path diff of interfering beams is } \Delta_{(air)} = DG - \mu EF$$

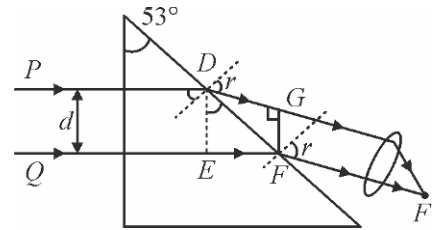
$$\text{Here } EF = d \tan \theta$$

$$\Delta_{(air)} = \mu d \tan \theta - \mu d \tan \theta = 0$$

$$DG = DF \sin \theta$$

$$= \frac{d}{\cos \theta} \cdot \sin \theta = \mu d \tan \theta \Rightarrow \text{height beams interfere at F constructively}$$

$$\Rightarrow I_F = (\sqrt{I_1} + \sqrt{I_2})^2 = (\sqrt{4I} + \sqrt{I})^2 = 9I$$



32. From figure we can use

$$\hat{R} - \hat{I} = 2 \cos \theta \hat{N}$$

$$\hat{R} = \hat{I} + 2 \cos \theta \hat{N}$$

$$\hat{R} = \left(\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}} \right) - \frac{2\sqrt{2}}{\sqrt{3}} \cdot \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$$

$$\hat{R} = -\frac{\hat{i}}{\sqrt{3}} - \frac{\hat{j}}{\sqrt{3}} - \frac{\hat{k}}{\sqrt{3}}$$

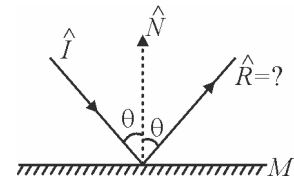
$$\hat{R} = -\frac{\hat{i}}{\sqrt{3}} - \frac{\hat{j}}{\sqrt{3}} - \frac{\hat{k}}{\sqrt{3}}$$

$$\hat{R} = -\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

$$\cos \theta = -\hat{I} \cdot \hat{N}$$

$$\cos \theta = -\frac{(\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} + \hat{j})}{\sqrt{3} \cdot \sqrt{2}}$$

$$\cos \theta = -\frac{2}{\sqrt{2}\sqrt{3}} = -\frac{\sqrt{2}}{3}$$



33. For mirror object distance is $4 = 15 + \frac{40}{4/3} = 45 \text{ cm}$

$$\text{Image distance is } v = 15 + \frac{16}{4/3} = 27 \text{ cm}$$

If f is the focal length of mirror, by mirror formula

$$f = \frac{\mu v}{\mu + v} = \frac{45 \times 27}{45 + 27} = \frac{45 \times 27}{72} = \frac{135}{8} = 16.875$$

34. For concave mirror $4 = -100 \text{ cm}$
 $f = -100 \text{ cm}$

By mirror formula
$$v = \frac{4f}{4-f} = \frac{-110 \times -100}{-110+100} = -1100 \text{ cm}$$

Final image is produced at a distance 100 cm below plane mirror.

35. For min deviation, we use- $\delta_1 = 2i - A$

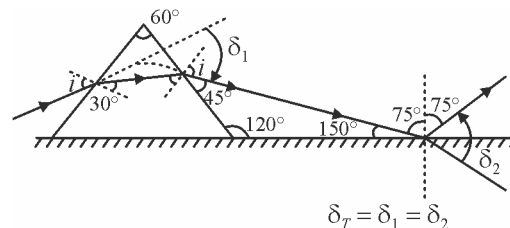
By Snell's law - $\sin i = \sqrt{2} \sin 30^\circ = \frac{1}{\sqrt{2}}$

$$i = 45^\circ$$

$$\Rightarrow \delta_1 = 90 - 60 = 30^\circ \text{ (clockwise)}$$

At plane mirror, deviation angle is $\delta_2 = 180 - 150 = 30^\circ \text{ (clockwise)}$

$$\delta_T = \delta_1 = \delta_2 = 0$$



36. By reflection/mirror formula, we use

$$\left[\begin{array}{l} 4 = -15 \text{ cm} \\ f = -10 \text{ cm} \end{array} \right] \Rightarrow v = \frac{4f}{4-f} = \frac{-15 \times -10}{-15+10} = \frac{+150}{-5} = -30 \text{ cm}$$

Magnification of image $m = \left| \frac{v}{u} \right| = \frac{30}{15} = 2$

Vel magnification $m^2 = 4$

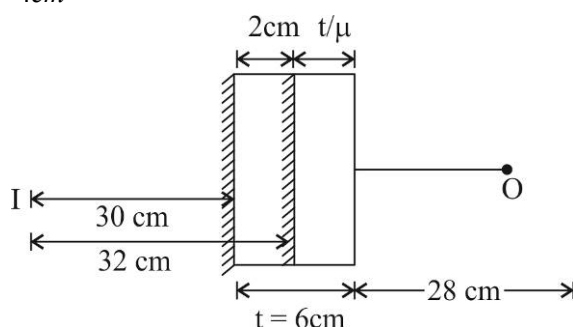
Vel of image w.r.t to mirror $u_{im} = m^2 \cdot v_{om} = 4 \times 11 = 44 \text{ m/s (right)}$

$$v_{im} = u_i - u_m = 44 \text{ m/s}$$

$$u_i = 44 + 1 = 45 \text{ m/s}$$

37. Distance of object from mirror is

$$28 \text{ cm} + \frac{6}{1.5} = 28 \text{ cm} + 4 \text{ cm} = 32 \text{ cm}$$



Final distance of image produced from structured face is

$$h = 32 \text{ cm} - 2 \text{ cm} = 30 \text{ cm}$$

38. At point of slope of trajectory of ray is $\frac{dy}{dx} = \cot \theta = \sqrt{y}$

$$\int_0^y \frac{dy}{\sqrt{y}} = \int_0^x dx$$

$$2\sqrt{y} = x \Rightarrow \boxed{y = \frac{x^2}{4}}$$

At $y = d = x = \sqrt{4d} = 2\sqrt{d}$

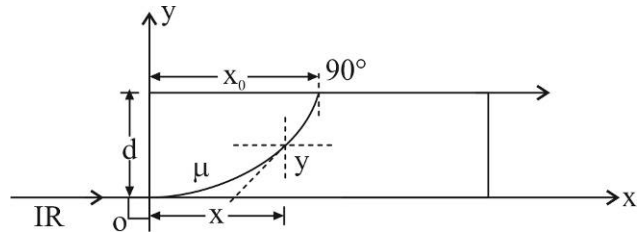
At pt. P, we use Snell's law

$$4 \sin \theta = 1$$

$$\sin \theta = \frac{1}{4}$$

$$\cos \theta = \sqrt{1 - \frac{1}{16}} = \sqrt{\frac{15}{16}} = \frac{\sqrt{15}}{4}$$

$$\cot \theta = \sqrt{15}$$



39. Slope of traj $\frac{dy}{dx} = \cot \theta = \sqrt{ay}$

$$\int_0^h \frac{dy}{\sqrt{ay}} = \int_0^\ell dx$$

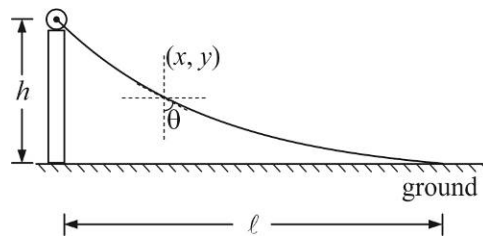
$$\left[2\sqrt{\frac{y}{a}} \right]_0^h = \ell \Rightarrow \ell = 2\sqrt{\frac{h}{a}}$$

By Snell's law

$$\mu_0 = \mu \sin \theta$$

$$\sin \theta = \frac{1}{\mu}$$

$$\cos \theta = \sqrt{1 - \frac{1}{\mu^2}} = \sqrt{\frac{\mu^2 - 1}{\mu^2}}$$



40. For min deviation, we use

$$\mu \sin \left(\frac{A}{2} \right) = \sin \left(\frac{A + \alpha}{2} \right)$$

$$\mu \sin \left(\frac{A}{2} \right) = \sin \frac{A}{2} \cos \frac{\alpha}{2} + \cos \frac{A}{2} \sin \frac{\alpha}{2}$$

$$\mu = \cos \left(\frac{ga}{2} \right) + \cot \left(\frac{A}{2} \right) \cdot \sin \frac{\alpha}{2}$$

$$\mu = \cos \frac{\alpha}{2} + \sin \frac{\alpha}{2}$$

.....(1)

By Snell's law, we use

$$\mu \cos \alpha = \sin e = \sin \beta$$

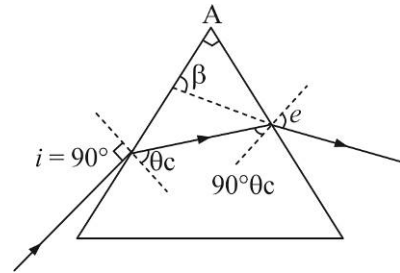
$$\mu \sqrt{1 - \frac{1}{\mu^2}} = \sin \beta$$

$$(\mu^2 - 1) = \sin^2 \beta$$

From equation (1)

$$\left(\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2} \right)^2 - 1 = \sin^2 \beta$$

$$\sin \alpha = \sin^2 \beta$$



41. Critical angle for glass is $\theta_c = \sin^{-1}\left(\frac{1}{\mu}\right) = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

For refraction we use

$$\frac{h_2}{v} - \frac{h_1}{v} = \frac{h_2 - h_1}{R}$$

$$\frac{2}{V} = \frac{1}{R}$$

$$\Rightarrow V = +2R$$

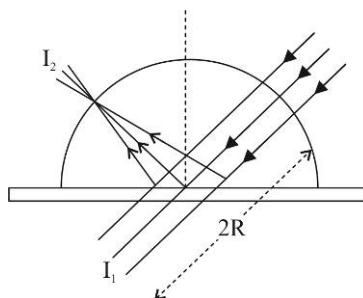
$$u = \infty$$

$$h_1 = 1$$

$$h_2 = 2$$

$$R = +R$$

$\Rightarrow$ Final image will be produced on the surface of hemisphere (I_2) as shown.



42. for achromatic doublet

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0 \quad \text{and} \quad \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{20} \quad \dots\dots(1)$$

$$\frac{0.03}{f_1} + \frac{0.05}{f_2} = 0 \quad \dots\dots(2)$$

From (1) & (2)

$$\frac{1}{f_1} = -\frac{5}{3f_2} \Rightarrow \frac{1}{f_2} - \frac{5}{3f_2} = \frac{1}{20}$$

$$-\frac{2}{3f_2} = \frac{1}{20} \Rightarrow f_2 = -\frac{40}{3} \text{ cm}$$

$$f_1 = 8 \text{ cm}$$

43. For lens $f = \frac{R}{2(\mu - 1)} \Rightarrow R = 2f(\mu - 1) = 2 \times 30 \times 0.5 = 30 \text{ cm}$

Ref from air to glass

$$u = -90 \text{ cm}; \mu_1 = 1$$

$$R = +30 \text{ cm}; \mu_2 = 3/2$$

$$\frac{\mu_2}{V} - \frac{\mu_1}{V} = \frac{h_2 - h_1}{R}$$

$$\frac{3}{2v} + \frac{1}{90} = \frac{y_2}{30}$$

For ref from glass to water

$$u = 270 \text{ cm}; \mu_1 = 3/2$$

$$R = -300 \text{ cm}; \mu_2 = 3/2$$

$$\frac{3}{2V} - \frac{4}{3 \times 40} = \frac{1/6}{-30}$$

$$\frac{3}{2V} = \frac{1}{180} + \frac{1}{180} V_2 = +120 \text{ cm}$$

For ref from water to glass

$$u = +40 \text{ cm}; \mu_1 = 4/3$$

$$R = -30 \text{ cm}; \mu_2 = 3/2$$

$$\frac{3}{2v} - \frac{4}{3 \times 30} = \frac{1}{-6}$$

$$\frac{3}{2v} = \frac{1}{30} - \frac{1}{180} = \frac{5}{180} = V_3 = +54 \text{ cm}$$

For ref from glass to air

$$u = +54 \text{ cm}; \mu_1 = 3/2$$

$$R = +30 \text{ cm}; \mu_2 = 1$$

$$\frac{1}{V} - \frac{3}{2 \times 54} = \frac{-y_2}{30}$$

$$\frac{1}{v} = \frac{1}{36} - \frac{1}{60} = \frac{1}{90} \Rightarrow v = +90 \text{ cm}$$

44. For πR to take place for all light rays entering at A we use

$$\theta \geq \theta_c$$

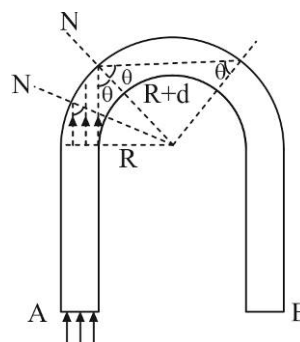
$$\sin \theta = \frac{1}{\mu} = \frac{R}{R+d}$$

$$\frac{R}{R+d} = \frac{2}{3}$$

$$3R = 2R + 2d$$

$$2d = R$$

$$\frac{d}{R} = \frac{1}{2}$$



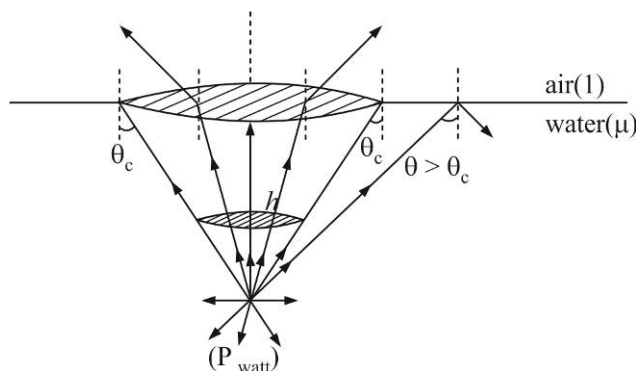
45. If P is source power which is isotropic in a solid angle 4π it is uniformly dist, the cone in which light escapes from center. Its solid angle is $\Omega = 2\pi(1 - \cos \theta_c)$

$$P_{es} = \frac{P}{4\pi} \cdot \Omega = \frac{P}{4\pi} \cdot 2\pi(1 - \cos \theta_c)$$

$$P_{es} = \frac{P}{2}(1 - \sqrt{1 - \sin^2 \theta_c})$$

$$P_{es} = \frac{P}{2} \left(1 - \sqrt{1 - \frac{1}{\mu^2}} \right)$$

$$f_{es} = \frac{P_{es}}{P} = \frac{1}{2} \left(1 - \sqrt{1 - \frac{1}{\mu^2}} \right)$$



46. In triangle CHN we have $\cos \theta = \frac{CN}{CB} = \frac{R/z}{CB}$

$$CB = \frac{R}{2 \cos \theta}$$

$$\text{At } \theta = 0 \quad CB = \frac{R}{Z} \text{ as } \cos \theta \rightarrow 1$$

$$BO = CB = f \text{ (Paraxial rays)}$$

47. We need to know how the acceleration due to gravity, g , varies with height above the Earth's surface. Taking the Earth to be spherically symmetric mass M , we can write

$$g = \frac{GM}{r^2}$$

Where r is the distance from the centre, assuming that r is greater than the Earth's radius. Differentiating both sides gives

$$dg = \frac{-2GM}{r^3} dr$$

Dividing by g (and substituting from our previous expression for g) gives

$$\frac{dg}{g} = -\frac{2dr}{r}$$

(which, apart from the $-$ sign, we recognize as the familiar result from error analysis). Thus the fractional change in r is half the fractional change in g (ignoring the minus sign), so the change in r required to achieve a fractional change of 6×10^{-9} in g must be

$$\frac{1}{2} \times 6 \times 10^{-9} \times 6.4 \times 10^6 \text{ m}$$

(we have taken the Earth's radius as 6400 km) = 0.019 m

I.e. change in height of 19 mm would cause a change in g of 6 parts in 10^9 .

To consider the effect of a change of latitude, we differentiate the expression $g = g_0(1 + \beta \sin^2 \phi)$ with respect to the latitude ϕ to obtain $g/d\phi = 2g_0\beta \sin \phi \cos \phi$. Thus

$$\Delta\phi = \frac{\Delta g}{2g_0\beta \sin \phi \cos \phi}$$

Taking $\Delta g/g = 6 \times 10^{-9}$, $\beta = 0.0053$ and $\phi = \pi/4$ radians gives $\Delta\phi = 1.13 \times 10^{-6}$ radians. Multiplying this by the Earth's radius of $6.4 \times 10^6 \text{ m}$ gives a distance on the Earth's surface of about 7 m.

48. (i) The fractional error in D^2 is $2(0.001/0.035) = 5.7\%$
 The fractional error in L^2 is $2(0.001/0.120) = 1.7\%$
 So the fractional error in D^2/L^2 is $(5.7^2 + 1.7^2)^{1/2} \% = 6.0\%$
 Calculating the value of D^2/L^2 gives 0.0851, so the absolute error in this quantity is 0.005.
 Thus $1 + D^2/L^2 = 1.085 \pm 0.005 = 1.085 \pm 0.5\%$
 So $(1 + D^2/L^2)^{1/2} = 1.042 \pm 0.25\%$.
 The fractional error in I is 1.0%, and we may assume that there is no error in the values of μ_0 or of n , so the fractional error in B must be $(0.25^2 + 1.0^2)^{1/2} = 1.0\%$. Thus (taking $\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$) we obtained
 $B = 9.08 \text{ mT} \pm 1.0\%$
 $9.08 \pm 0.09 \text{ mT}$.
- (ii) Fractional error in V is $1/20 = 5.0\%$
 Fractional error in r^2 is $2(0.1/3.4) = 5.9\%$
 Fractional error in B^2 is $2 \times 1.08\% = 2.0\%$
 So fractional error in e/m is $(5.0^2 + 5.9^2 + 2.0^2)^{1/2} \% = 8.0\%$
 Thus $e/m = 1.68 \times 10^{11} \text{ Ckg}^{-1} \pm 8.0\% = (1.7 \pm 0.1) \times 10^{11} \text{ Ckg}^{-1}$.
 We can compare this with the currently accepted value for the charge-to-mass ratio of the electron, which is $1.759 \times 10^{11} \text{ Ckg}^{-1}$. It is clear that the experimental value is consistent with this value of within the experimental error.

49. Smallest main scale division = 0.05 cm (5.15 – 5.10)
Main scale reading = 5.10 cm (comes just before the zero of the Vernier scale)
Diameter
= Main Scale Reading + Vernier coincidence $\times$ Least Count
= 5.10 + 24 $\times$ 0.001 = 5.124 cm
50. $MSD = 1/10 \text{ cm} = 0.1 \text{ cm}$
For first vernier caliper, 10 $VSD = 9 \text{ MSD}$
 $\Rightarrow VSD = 9/10 \text{ MSD} = 0.9 \times 0.1 = 0.09 \text{ cm}$
Reading = main scale reading upto coinciding main scale division $-n \times VSD$
= 3.5 – 7 $\times$ 0.09 = 3.5 – 0.63 = 2.87 cm
For second vernier caliper, 10 $VSD = 11 \text{ MSD}$
 $\Rightarrow VSD = 11/10 \text{ MSD} = 1.1 \times 0.1 = 0.11 \text{ cm}$
Reading = 3.6 – 7 $\times$ 0.11 = 3.6 – 0.77 = 2.83 cm
51. Least count = pitch / no. of division on circular scale = 0.5 mm / 50 = 0.01 mm
Negative zero error = $-5 \times LC = -5 \times 0.01 = -0.05 \text{ mm}$
Measured value = main scale reading + screw gauge
Reading – zero error
= 0.5 mm + 25 $\times$ 0.01 mm – (–0.05 mm) = 0.75 mm + 0.05 mm = 0.80 mm
52. Vernier caliper :
 $M.S.D = 1/8 \text{ cm} = 0.125 \text{ cm} = 1.25 \text{ mm}$
 $5 \text{ V.S.D.} = 4 \text{ M.S.D.}$
 $\Rightarrow V.S.D. = 4/5 \text{ M.S.D.} = 4/5 \times 1.25 \text{ mm} = 1 \text{ mm}$
Least count = $M.S.D. - V.S.D. = 1.25 - 1 = 0.25 \text{ mm}$
Screw gauge : No. of circular scale divisions = 100
1 rotation (pitch) = 2 linear scale divisions
If the pitch of the screw gauge is twice the least count of the Vernier calipers, the least count of the screw gauge is
Pitch = 2 $\times$ Least Count of vernier calipers = 2 $\times$ 0.25 mm = 0.5 mm
Least count of the screw gauge = Pitch / no. of circular scale divisions = 0.5 / 100 = 0.005 mm
Option (B) is correct
If the least count of the linear scale of the screw gauge is twice the least count of the Vernier calipers, the least count of the screw gauge is
Linear scale division of screw gauge
= 2 $\times$ Least count of vernier calipers
= 2 $\times$ 0.25 mm
= 0.5 mm
Now,
Pitch = 2 linear scale divisions = 2 $\times$ 0.5 = 1 mm
Least count of the screw gauge
= Pitch / No. of circular divisions = 1/100 mm = 0.01 mm
Option (C) is correct